UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

ASSIGNMENT 2

DUE DATE FOR ASSIGNMENT 2: WEEK 12, SUNDAY BEFORE 11:55PM

Please READ and FOLLOW all the INSTRUCTIONS carefully. You can reuse the code covered in the lectures or from the textbook.

1. EQUIPMENT

Hardware: Any suitable machine, as long as it allows for a complete assignment.
Software: Microsoft Visual Studio 2026 Community Edition. The submitted program will be tested by the markers using Visual Studio C++ or other software.

2. PROJECT:

This is a **GROUP** project with **THREE** members.

3. DESCRIPTION OF PROJECT

The assignment focuses on some important aspects of structured programming practices. The aim of this assignment is to provide an opportunity to demonstrate that you have acquired the skills and ability to be proficient in the area of developing C++ application program using modular approach.

4. ACADEMIC HONESTY AND COLLABORATION

Cooperation is recommended ONLY in understanding various concepts and system features. However, the actual solution of the assignments, the programming and debugging must be your individual work, except for what you specifically credit to other sources. (Your grade will be based on your own contribution.) For example, copying without attribution any part of someone else's program is plagiarism, even if you modify it and even if the source is a textbook. You can document the credit to other sources at the start of your program code listing. Offenders will be awarded with **ZERO** mark for this assignment.

5. PENALTY

Penalties such as downgrade or reduction of marks will be given to empty or late submissions unless legitimate reasons are provided.

Descriptions

I. It is a group assignment with **3 members** (except special cases).

II. Given a scenario:

NextFlip is a subscription-based video-on-demand streaming service that distributes both original and licensed films and television shows across various genres. The platform is available internationally and supports multiple languages. Users can add or remove their favourite titles from a personalised list. In addition, merchandise is offered within the platform, allowing users to make purchases using the credit available in their accounts. A system will be developed to integrate and support all of these user functions.

III. **Each team member is required to choose and develop three/four specific modules as listed** below:

Table 1: Sub-functions (sub-modules) of the program

Function (module)	Details
Member 1 (Role: **COMBINE CODE**) **– LOGIN** modules	
loadUserInfo	A function to read all the registered users' details from "userInfo.txt"
registerNewUser	A function to register new user and save it to "userInfo.txt"
login	A function to validate both the username and password, ensuring secure user login access to the program.
Member 2 (Role: **FIX ERRORS**) – **FAVOURITE** modules	
loadFavInfo	A function to read the favourite(s) saved in "XXX MDInfo.txt"
addFavList	A function to add favourite to the list and update the "XX MDInfo.txt"
displayFavList	A function to display all the favourite(s) stored in the text file ("XXX MDInfo.txt")
deleteFavList	A function to delete selected favourite in the list and update the "XX MDInfo.txt"
Member 3 (Role: **ERROR CHECKING**) – **MERCHANDISE** modules	
loadPurchaseInfo	A function to read the credit and merchandise saved in "XXX PurInfo.txt"
addItem	A function to allow all users to easily add credits to their accounts, with corresponding records being updated for transparent transaction tracking. Besides, it also allows all users to add the merchandise to their accounts. All the information is saved (updated) to "XXX PurInfo.txt"
payment	A function to make payment for the merchandise
printReceipt	A function to print the receipt of purchase for the merchandise "XXX Receipt.txt"

IV. The project consists of **at least 11 modules** and it is **MANDATORY** to adhere to the structure outlined in both the provided template and the designated text file for data storage. The group must use the given template and text files to kickstart the program during implementation – **no other template structure or text file structure is allowed (modification of text file structure is definitely prohibited!)**.

V. It is crucial to **implement** the program using the **provided coding template** and carefully follow the instructions outlined within it. Cover all the topics except Struct (Introduction till String). Marks will be deducted if any of the topics mentioned such as IO, array, function or others are not covered.

VI. It is important that the flow of the whole program be smooth and work logically like a real program. Each team member is assigned with a specific role:- 1) Combine code, 2) Error checking and, 3) Fix errors. **The roles can be rotated among team members** (instead of following the assigned roles in Table 1), and it is **important to clearly state any role transitions** in the report. The responsibilities include:

1. Combine Code: Integrate all modules to form a complete executable program after individual implementation.
2. Error Checking: Inspect for syntax, semantic, and runtime errors.

3. Fix Errors: Address and rectify all identified bugs, ensuring the program operates seamlessly.

VII. While specific roles are assigned, it is crucial for all members to collaborate both collectively and individually to successfully accomplish all the program modules.

VIII. **Creative decorations** can be included in the program without interfering with the overall structure.

MORE INSTRUCTIONS

1. Assignment's Instructions:

- This is a **group assignment**.
- It is recommended to read this assignment several times before coding.
- Cover the lecture topics **except Struct** (**Introduction till String**).
- Please take note that **your submitted assignment must be within the course scopes**. Any submission which is **OUT OF THE TOPICS** (such as Class, Linked List, Pointer,…) will be awarded with **ZERO MARK**.

2. Report Submission

- The **REPORT** should **include** the following items:

	Item	Description	Tick
i	Cover page	Group Number, name of members, student ID, programme name (e.g., Bachelor of Computer Science), course code and name (to make sure we receive the correct assignment), and email address. **(*A sample is provided at the last page)**	☐
ii	Task Division	Specify the modules completed by the members **(*A sample is provided at the last page)**	☐
iii	Objective	Objective of the A2 (objective of the program instead of the objective of the subject)	☐
iv	Pseudocode	1. Write a pseudocode for the main function (it should not be written in C++ code). 2. Write the pseudocode for the sub-functions outlined in Table 1. 3. Ensure that the pseudocode is sufficiently generalised, making it accessible even to individuals unfamiliar with programming concepts.	☐
v	Flowchart	1. Draw flowchart for the main function 2. Draw the flowchart for the sub-functions outlined in Table 1. 3. Ensure that the flowchart is sufficiently generalised, making it accessible even to individuals unfamiliar with programming concepts.	☐
vi	Test cases	Screenshot of the program output. Please use 2 to 4 different inputs for each member. Each test case should address a particular feature. Note that your test cases should be consistent with your chosen objective.	☐
vii	Source code	Source code, which is exactly the same as the softcopy submitted to WBLE	☐

- **ZIP** the **SOFTCOPY** of your ***.cpp file, *.exe file and** the **REPORT (.pdf file) together and SUBMIT the .zip file** to WBLE before the due date, which is posted on WBLE. The text files are not needed.
- No idea about zip file? Please refer to google or Wikipedia (https://en.wikipedia.org/wiki/ZIP_(file_format)) or 7zip (https://www.7-zip.org/).
- The **.exe** file is very **IMPORTANT** and must be included into the zip file.
- Remember to rename all the files (*.zip, *.cpp, *.exe, *.pdf) to your group number as filled in the google sheet, E.g. Group32.exe. **NOTE: ONLY ONE** member will submit the files.
- No idea about **.exe file?** Please refer to practical 1.
- The report is not expected to use more than 120 pages. Set single line spacing, font size 11 and choose the font type Times New Roman (Consolas for codes). Exceeding pages will not be reviewed.

3. Suggest Grading Scheme

- Correctness of the program according to the provided template.
- Report structure: completeness and quality of presentation of each item.
- Documentation of codes (how easy to understand your codes); However, avoid excessive comments, such as commenting every line of your code.
- Total marks for Assignment 2 are 60 marks (program 45 + report 15) and contributes 15% to the final grade;
- The marking scheme shown below is just a **GUIDELINE** to develop the complete program and also report;
- The marks may be adjusted according to various circumstances;
- The penalties will be complied as described below:

Suggested Marks		
	Details	**Suggested Grading Range**
1	Compilation (able to compile)	0 - 10
	Execute (able to execute)	
	Comments (precise comments provided)	
	Input validations (important validations covered)	
2	Modules (according to modules in charge, evaluated individually)	0 - 18
3	Effort (decoration, …..)	0 - 6
4	Overall (completeness of the program)	0 - 5
5	Overall contribution of each member (roles)	0 - 6
	Subtotal marks (program)	45
	Subtotal marks (report)	15
	Grand Total	60
	Penalties	
1	Late submission **OR** wrong submission (re-submission)	**10% deducted per day from total marks that earned**
2	**Out of topics** ✓ Class, Linked List, Pointer, Enum, …. ✓ Other libraries that have not been mentioned in the UCCD1004 learning materials are prohibited **UNLESS** it is for decoration purposes (put remark to state it clearly) ✓ E.g. using #include <array> is prohibited	**-45**

Example Program as a Reference

Member 1: Register new user and login

1. When the user is not registered, they are required to complete the registration process by providing a name, email, username, and password (without spaces) (as illustrated in Figure 1). The details of the newly registered user will be saved (appended) to the text file "userInfo.txt", following the format depicted in Figure 2.

```
1. Register New User
2. Login
3. Quit
Choice: 1
Name: Taylor Lee Xiao Long
Email: taylorlee@gmail.com
Username: taylor
Password (At least 8 characters (1 symbol, 1 uppercase letter, 1 number)): ~Taylor123
Confirm registration? (1-yes, 2-no): 1
```

Figure 1: Register a new user

Figure 2: Save the details of the newly registered user into the "userInfo.txt".
Left: "userInfo.txt" before a new user register, **Right**: "userInfo.txt" after a new user registered

"userInfo.txt" is essential for storing the user details and its text file structure should adhere to the following format:

```
[name with space]
[email]
[username]
[password]

E.g.,
Rebecca Chin ← name with space
rebeccachin@gmail.com ← email
rebecca ← username
#Rebecca2012 ← password
```

2. The system needs to check the format of the email to make sure the correct email is provided by the user as shown in Figure 3.

```
1. Register New User
2. Login
3. Quit
Choice: 1
Name: Taylor Lee Xiao Long
Email: taylorlee
Please key in the correct email!
Email:
```

Figure 3: The "@" symbol is missing when the user enters the email address; therefore, a reminder pops up to prompt the user to provide a valid email address.

3. In addition, the system must ensure that users provide a password in the correct format, with at least 8 characters, including one symbol, one uppercase letter, and one number, as shown in Figure 4.

```
1. Register New User
2. Login
3. Quit
Choice: 1
Name: Taylor Lee Xiao Long
Email: taylorlee
Please key in the correct email!
Email: taylorlee@gmail.com
Username: taylor
Password (At least 8 characters (1 symbol, 1 uppercase letter, 1 number)): sdf
The length of the password is less than 8. Please key in again.
Password (At least 8 characters (1 symbol, 1 uppercase letter, 1 number)):
```

```
1. Register New User
2. Login
3. Quit
Choice: 1
Name: Taylor Lee Xiao Long
Email: taylorlee
Please key in the correct email!
Email: taylorlee@gmail.com
Username: taylor
Password (At least 8 characters (1 symbol, 1 uppercase letter, 1 number)): sdf
The length of the password is less than 8. Please key in again.
Password (At least 8 characters (1 symbol, 1 uppercase letter, 1 number)): ~Taylor123
Confirm registration? (1-yes, 2-no): 1
Do you want to continue register/login? (1-yes, 2-no):
```

Figure 4: **Top**: A reminder pops up to prompt the user to provide a valid password; **Bottom**: Registration is confirmed when all inputs are valid.

4. The user can successfully login when the entered username and password are matched. Upon successful login, the name is prominently displayed on the menu page, as depicted in Figure 5. In cases where the username is not found or the password is incorrect, appropriate messages are displayed, as shown by the example in Figure 6.

```
1. Register New User
2. Login
3. Quit
Choice: 2
Username: taylor
Password: ~Taylor123
Login Successfully!
Press any key to continue . . .
```

```
Name: Taylor Lee Xiao Long
~~Welcome to NextFlip~~
A. Favourite List
B. Purchase Merchandise
C. Exit
Choice:
```

Figure 5: Left: Able to login with the correct username and password, Right: Name is displayed on the menu page.

```
1. Register New User
2. Login
3. Quit
Choice: 2
Username: taylor
Password: ~asdf12
Username or Password not correct!
If you have not registered, please register first!
Do you want to continue register/login? (1-yes, 2-no):
```

Figure 6: Displays the message when the username or password is not correct

Member 2: Add new favourite, delete favourite and display the favourite(s)

1. User need to select 'a' to go into the "Favourite" section as shown in Figure 7.

```
Name: Taylor Lee Xiao Long
~~Welcome to NextFlip~~
A. Favourite List
B. Purchase Merchandise
C. Exit
Choice: a
```

Figure 7: Select 'a' to access the "Favourite List" and select 'b' to access "Purchase Merchandise."

2. In favourite section, menu to display favourite, add favourite and delete favourite are displayed as shown in Figure 8. If a **new user** selects option '1', a message such as "There is no favourite in the list!" will likely be displayed. This occurs when a new user registers and no favourite is added to the list yet. This might also happen when the user deletes all the favourite(s) in the list.

```
1. Display Favourite List
2. Add Favourite List
3. Delete Favourite List
4. Back
Select: 1
There is no favourite in the list!
Press any key to continue . . .
```

Figure 8: Selection of menu: Select '1' to display all the favourite(s) in the list. "There is no favourite in the list!" is displayed when the favourite is empty.

3. Option '2' is selected to add a new favourite to the list. Details such as category, title, year, language, description, and rating must be entered in order to add the item to the favourite list as shown in Figure 9.

```
1. Display Favourite List
2. Add Favourite List
3. Delete Favourite List
4. Back
Select: 2
Add New Favourite Drama or Movie:
Category: Drama
Title: I Fight with A Dragon
Year: 2026
Language: Chinese
Description: Romance, Violence
Rating: 9.0
Confirm add? (1-yes, 2-no): 1
```

Figure 9: Select '2' and the details of the favourite are keyed in by the user.

Once a new favourite is added into the list, it should be updated in the “XX MDInfo.txt” as shown in Figure 10.

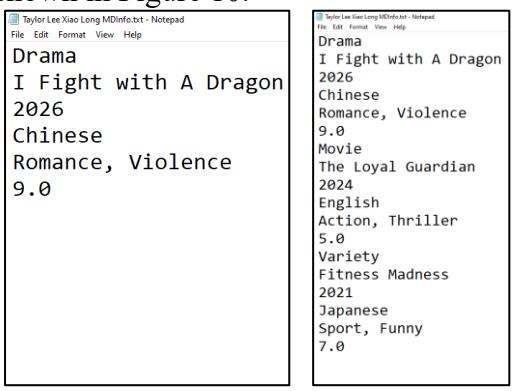

Figure 10: New favourite is added by the user and “XX MDInfo.txt” should be updated accordingly. **Left**: A new favourite is added and updated in the text file; **Right**: More than one favourite are added into the list and updated in the text file. All the favourite(s) add by the particular user are stored in “XX MDInfo.txt”.

“XX MDInfo.txt” is essential for storing the favourite keyed in by the respective user and its text file structure should adhere to the following format:

```
[category]
[title]
[year]
[language]
[descriptions]
[rating]

E.g.,
Drama ← category
I Fight with A Dragon ← title
2026 ← year
Chinese ← language
Romance, Violence ← descriptions
9.0 ← rating
```

E.g., when Taylor Lee Xiao Long login and adds a Chinese romance (and violence) drama titled “I Fight with a Dragon”, released in 2026 with a rating of 9.0, the details of this drama are recorded in a custom text file named “Taylor Lee Xiao Long MDInfo.txt.” The variable “XX” is dynamically tailored to match the login name.

4. All the favourite(s) will be displayed when the user selects option ‘1’ as shown in Figure 11.

```
1.
Category: Drama
Title: I Fight with A Dragon
Year: 2026
Language: Chinese
Description: Romance, Violence
Rating: 9.0
2.
Category: Movie
Title: The Loyal Guardian
Year: 2024
Language: English
Description: Action, Thriller
Rating: 5.0
3.
Category: Variety
Title: Fitness Madness
Year: 2021
Language: Japanese
Description: Sport, Funny
Rating: 7.0
Press any key to continue . . .
```

Figure 11: All the favourite(s) are displayed.

5. When the user chooses option ‘3’, the titles of all the favourite(s) are displayed, and the user can select a favourite to delete. The selected favourite is then removed, and the file “XX MDInfo.txt” is updated accordingly by deleting the corresponding entry as shown in Figure 12.

```
1. Display Favourite List
2. Add Favourite List
3. Delete Favourite List
4. Back
Select: 3
1. I Fight with A Dragon
2. The Loyal Guardian
3. Fitness Madness
Delete: 2
Confirm Delete? (1-yes, 2-no): 1
```

Taylor Lee Xiao Long MDInfo.txt - Notepad
File Edit Format View Help

```
Drama
I Fight with A Dragon
2026
Chinese
Romance, Violence
9.0
Variety
Fitness Madness
2021
Japanese
Sport, Funny
7.0
```

Figure 12: **Left**: The user selects ‘2’ to delete a favourite titled “The Loyal Guardian”; **Right**: “The Loyal Guardian” is removed from “XX MDInfo.txt” as it has been selected for deletion.

Member 3: Add credit, add merchandise and make payment

1. User needs to select 'b' to go into the "Merchandise" section as shown in Figure 13.

```
Name: Taylor Lee Xiao Long
~~Welcome to NextFlip~~
A. Favourite List
B. Purchase Merchandise
C. Exit
Choice: b
```

Figure 13: Select 'a' to access the "Favourite List" and select 'b' to access "Purchase Merchandise."

2. In the merchandise section, the user can choose to add credit or add merchandise or make a payment as shown in Figure 14. If user select '1', it will show the current credit available in the user account based on the information stored in "XX PurInfo.txt".

```
1. Add Credit/Merchandise
2. Make Payment
3. Back
Choice: 1
```

```
Current Credit: RM 0.00
1. Add Credit
2. Add Merchandise
3. Back
Choice:
```

Figure 14: **Left**: Options to add credit, add merchandise, or make a payment in the merchandise section; **Right**: The current credit is displayed along with the options to add credit or add merchandise.

If the current user is new, the credit will be zero and there will be no merchandise listed in the "XX PurInfo.txt" as shown in Figure 15.

```
Taylor Lee Xiao Long PurInfo.txt - Notepad
File  Edit  Format  View  Help
0
```

Figure 15: "XX PurInfo.txt" is used to store the credit and the merchandise available in the user's account.

3. If "Add Credit" is selected, the user will enter the amount of credit to be added to the account, as shown in Figure 16. This credit is then used to pay for merchandise added to the account.

```
Current Credit: RM 0.00
Credit to add: RM 1000
Confirm? (1-yes, 2-no): 1
```

```
Current Credit: RM 1000.00
1. Add Credit
2. Add Merchandise
3. Back
Choice:
```

Figure 16: **Left**: The user chooses to add RM 1000 to the account; **Right**: The added RM 1000 credit is reflected in the menu once the addition is confirmed.

4. If “Add Merchandise” is selected, the user needs to enter the name of the merchandise, the quantity to purchase, and the price per piece. All previously entered merchandise will be listed whenever the user adds new merchandise to the list, as shown in Figure 17.

```
Current Credit: RM 1000.00

~Add Merchandise~
Name: The Loyal Guardian Socks
Quantity: 5
Price (per piece): RM 25
Confirm? (1-yes, 2-no): 1
```

```
Current Credit: RM 1000.00
~Item 1~
Name: The Loyal Guardian Socks
Quantity: 5
Price (per item): RM 25.00
~Item 2~
Name: Only Me T-shirt
Quantity: 10
Price (per item): RM 35.00

~Add Merchandise~
Name: Cooking Madness Frying Pan
Quantity: 1
Price (per piece): RM 300
Confirm? (1-yes, 2-no): 1
```

Figure 17: **Left**: the 1st item is added to the list; **Right**: All previously entered items are listed whenever the user adds new merchandise.

All the added items (credit and merchandise) are saved to the respective text file “XX PurInfo.txt” as shown in Figure 18.

```
Taylor Lee Xiao Long PurInfo.txt - Notepad
File  Edit  Format  View  Help
1000
The Loyal Guardian Socks
5
25
Only Me T-shirt
10
35
Cooking Madness Frying Pan
1
300
```

```
[credit]
[name of merchandise]
[quantity]
[price per piece]
......

1000 ← credit
The Loyal Guardian Socks ← name
5 ← quantity
25 ← price per piece
Only Me T-shirt
10
35
Cooking Madness Frying Pan
1
300
```

1st item

2nd item

3rd item

Figure 18: **Left**: The user’s credit balance and added merchandise are saved in “XX PurInfo.txt”; **Right**: The storage format for the credit and merchandise. The first line indicates the available credit in the account, followed by the details of the merchandise added by the user.

“XX PurInfo.txt” is essential for storing the available credit and the merchandise added by every respective user, and its file structure must adhere to the format shown in Figure 18. For example, when Taylor Lee Xiao Long logs in and adds RM 1000 in credit along with three different merchandise items, this information is saved in “Taylor Lee Xiao Long PurInfo.txt.”

5. When “Make Payment” is selected, all merchandise items are displayed, and the total price is calculated and shown on the screen as shown in Figure 19.

```
1. Add Credit/Merchandise
2. Make Payment
3. Back
Choice: 2
Current Credit: RM 1000.00
~Item 1~
Name: The Loyal Guardian Socks
Quantity: 5
Price (per piece): RM 25.00
~Item 2~
Name: Only Me T-shirt
Quantity: 10
Price (per piece): RM 35.00
~Item 3~
Name: Cooking Madness Frying Pan
Quantity: 1
Price (per piece): RM 300.00
Total Price (All):
(5 * RM 25.00) + (10 * RM 35.00) + (1 * RM 300.00) = RM 775.00
Confirm payment? (1-yes, 2-no): 1
Complete payment! Please check the receipt for details!
Press any key to continue . . .
```

Figure 19: All the merchandise and the total price are displayed when the user decides to make the payment.

6. A receipt, “XX Receipt.txt” will be generated when the payment is made as proof of payment, as shown in Figure 20. The remaining credit, after the purchase deduction, is also indicated on the receipt.

Taylor Lee Xiao Long Receipt.txt - Notepad
File Edit Format View Help

```
Receipt of Purchase
~Item 1~
Name: The Loyal Guardian Socks
Quantity: 5
Price (per item): RM 25.00
~Item 2~
Name: Only Me T-shirt
Quantity: 10
Price (per item): RM 35.00
~Item 3~
Name: Cooking Madness Frying Pan
Quantity: 1
Price (per item): RM 300.00
Total Price (All):
(5 * RM 25.00) + (10 * RM 35.00) + (1 * RM 300.00) = RM 775.00
Available Credit: RM 1000.00
Remain Credit: RM 1000.00 - RM 775.00 = RM 225.00
```

Figure 20: Receipt of the payment is generated when the payment is made.

Please note that “XX PurInfo.txt” should be updated after the payment is completed. The remaining credit must be updated, and the merchandise list should be cleared, as the payment has been processed, as shown in Figure 21.

Taylor Lee Xiao Long PurInfo.txt - Notepad
File Edit Format View Help

```
225
```

Figure 21: The remaining credit is RM225 after the last payment, and the merchandise list in “XX PurInfo.txt” is now empty.

7. Please take note that payment will be denied if the current credit is insufficient, as shown in Figure 22.

```
1. Add Credit/Merchandise
2. Make Payment
3. Back
Choice: 2
Current Credit: RM 225.00
~Item 1~
Name: Tiger Mama Mascot
Quantity: 1
Price (per piece): RM 300.00
Total Price (All):
(1 * RM 300.00) = RM 300.00
Confirm payment? (1-yes, 2-no): 1
Credit is insufficient!
Press any key to continue . . .
```

Figure 22: Payment is denied because the current credit is RM225, while the total merchandise cost is RM300. The available credit is insufficient to complete the payment.

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ACADEMIC YEAR 2026/2027

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UCCD 1004 PROGRAMMING CONCEPTS AND PRACTICES

ASSIGNMENT 2

Group XX			
Name	**ID**	**Programme**	**UTAR Email**

Task Division

	Name	Role	Modules
1.			
2.			
3.			